



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS FALL 2022

Faculty	FC&AS	Department	CS&IT	Batch	2020F
Semester	5 th	Degree Program	BSCS	Course Instructor	Ms. Ayeasha Urooj, Dr. Anam Akbar
Course Code	CS-310	Course Title	Compiler Construction	Credit Hours	3+0
Date of Examinations	02-02-2023	Timings	9:30 to 12:30	Shift	Morning

Timings: 3 Hours

Max Marks: 50

Instructions:

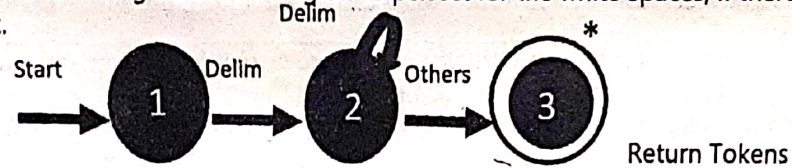
- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.
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Roll No.		Name		Section		'A' Copy Serial No.	
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Q.No.1**(2+2+2+2+2)**

- (a) Make a Transition diagram for all the signed numbers and also write the regular definition for all the signed numbers

- (b) Identify that the following transition diagram is perfect for the white spaces, if there is any error then justify it.



- (c) Panic Mode recovery: On discovery of an error, a parser may perform, local correction on the remaining input that is it may replace a Prefix of the remaining input by some string that allows the parse to continue.(T/F)(Justify your answer.)
- (d) Write the type expression:
An array of pointer to int, when the array index ranges from 1 to 10.



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(e) Type checking of statements are appropriate

- | | | |
|-----|--------------------------------------|---|
| i) | $T \rightarrow \text{char}$ | $\{T.\text{type} := \text{char}\}$ |
| | $T \rightarrow \text{integer}$ | $\{T.\text{type} := \text{integer}\}$ |
| | $T \rightarrow \uparrow T_1$ | $\{T.\text{type} := (T_1.\text{Type})\}$ |
| | $T \rightarrow \text{array}$ | $\{T.\text{type} := \text{array}(1 \dots \text{num.val}, T_1.\text{Type})\}$ |
| ii) | $E \rightarrow E_1 \text{ mod } E_2$ | $\{ E_1.\text{type} := \text{if } E_1.\text{Type} = \text{str and}$
$E_2.\text{type} = \text{str then str}$
$\text{Else type_error}\}$ |

Q.No.2

(10)

Find **FIRST()** and **FOLLOW()** of the given grammar and prove that grammar is LL(1)

$Q \rightarrow (P) | b$

$P \rightarrow P, Q | Q$

Q.No.3

(10)

Make SLR(1) Parser for the given Grammar

$Q \rightarrow Q + P | P$

$P \rightarrow P B | B$

$B \rightarrow B * | q | p$

Q.No.4

(5+5)

a) Refer grammar of Q: no:03 to perform stack implementation for the input string: qp^*q+p

b) Refer grammar of Q: no:02 to perform stack implementation for the input string: $(b, (b, b))$

Q.No.5

(5+5)

- a) Write the appropriate grammatical clauses with production rule and construct the dependency graph for the given input string: **real id1+id2*id3**
- b) Write the appropriate grammatical clauses, with its semantic rules for the input string: **a+c-9**, further construct its DAG.



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Faculty	FoC&AS	Department	CS&IT	Batch	2020F
Semester	5 th	Degree Program	BS(CS)	Course Instructor	Ahsan Masroor/Mubeen
Course Code	MS301:	Course Title	Probability & Statistics	Credit Hours	3+0
Date of Examinations	09-02-2023	Timings	9:30 AM – 12:30 PM	Shift	1 st

Timings: 3 Hours

Max Marks: 50

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Q.No.1 (5+5)

- (a) It is known from past records that 10% of the passengers arriving at Islamabad international airports are not properly vaccinated against yellow fever. 8 passengers are randomly selected to check whether they are vaccinated. What is the probability that 2 are not vaccinated?
- (b) Suppose that a secretary makes 3 errors per page on the average. What is the probability that on the next page she makes. i) No errors ii) 5 errors.

Q.No.2 (5+5)

- (a) A machine fills bottles of some soft drink. Two percent of the time it under fills and one percent of the time it overfills. What is the probability that a randomly selected bottle is either overfilled or underfilled?
- (b) A firm has two vacancies to fill is trying to recruit recent graduates. In the past, 40% of the candidates who were offered the job have not reported to join. If 2 candidates are offered job, what is the probability that at least one will join?

Q.No.3 (4+4+2)

Let us examine the case of an annual Ministry of Transport test to which all cars, irrespective of age, have to be submitted. The test looks for faulty brakes, steering, lights and suspension, and it is discovered after the first year that approximately the same numbers of cars have 0, 4, 8, 12 faults. The above situation is equivalent to drawing all



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possible samples of size 2 from this probability distribution (i.e. the population) **WITHOUT REPLACEMENT.**

- Verify that mean of all sample means is equal to population mean
- Find out standard deviation of population and sampling
- Determine standard error.

Q.No.4

(10)

A machine fills milk bottles, the means amount of milk in each bottle is supposed to be 250ml. In a routine check to see that the machine is operating properly, 36 filled bottles are chosen at random and found to contain a mean of 251.16 ml with a standard deviation of 2.4 ml. At a 0.05 level of significance, is the machine operating properly? Using the following table.

Test	α	0.01	0.02	0.05	0.10
Two tail	$Z_{\alpha/2}$	2.576	2.33	1.96	1.645
One tail	Z_{α}	2.33	2.05	1.645	1.28

Q.No.5

(3+3+2+2)

Data on age and price for a sample of 11 Orions in the table given below:

- Determine the regression equation for the data.
- Visualize regression equation and the data points.
- Describe the apparent relationship between age and price of Orions.
- Interpret the slope of the regression line in terms of prices for Orions.

Age (yr) X	5	4	6	5	5	5	6	6	2	7	7
Price (\$ 100) Y	85	103	70	82	89	98	66	95	169	70	48



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END SEMESTER EXAMINATIONS FALL 2022

Faculty	Faculty of Computing and Applied Sciences	Department	CS&IT	Batch	2020F
Semester	5 th	Degree Program	BS (Computer Science)	Course Instructor	Engr. Muhammad Nadeem
Course Code	IT-315	Course Title	Enterprise Systems	Credit Hours	3+0
Date of Examinations	11-Feb-2023	Timings	2:00 - 5:00	Shift	Evening

Timings: 3 Hours

Max Marks: 50

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Q.No.1**08 Marks**

The return on investment (ROI) of Enterprise resource planning (ERP) is crucial to building a business case that justifies an ERP investment in the first place. Further, the analysis that goes into quantifying the ROI of ERP helps businesses evaluate and anticipate the impact of an ERP system on their organizations and also refine their ERP approach over time so that the ERP's value to the business continues to increase. Briefly explain the key ERP ROI Measurement Parameters

Q.No.2**08 Marks**

The Human Resources (HR) module is a crucial component of most enterprise systems. It covers all activities dealing with the management of people in an organization. Discuss the core business functions of Human Resources (HR) module.



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Q.No.3

08 Marks

Procurement is the process of purchasing the goods and services that a company needs to support its operations. Procurement plays a vital function in every business. Operations will come to a standstill without the necessary goods and services a company needs. Illustrate the procurement cycle with brief explanation.

Q.No.4

08 Marks

Decision support system (DSS) is a computer-based information system that organizes, collects and analyzes business data that can be used in decision-making activities for the management, operation and planning in an organization or business. Describe the two real world example of DSS in industry and list down the tools available in Microsoft Excel for it.

Q.No.5

08 Marks

Business Process Reengineering (BPR) involves the radical redesign of core business processes to achieve dramatic improvements in productivity, cycle times and quality. Explain the steps involve in implementation of BPR in Organization.

Q.No.6

10 Marks

Data warehouse (DWH) is a repository where an organization electronically stores data by extracting it from operational systems, and making it available for ad-hoc queries and scheduled reporting. Illustrate how data warehouse works and describe its architecture?



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Faculty	Computing & Applied Sciences	Department	CS&IT	Batch	2020F
Semester	5th	Degree Program	BS(CS)	Course Instructor	Razia Nisar/ Tehmeed
Course Code	CS-317T	Course Title	Computer Networks	Credit Hours	3+1
Date of Examinations	13-02-23	Timings	2.00 pm to 5.00 pm	Shift	Evening

Timings: 3 Hours

Max Marks: 50

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Q.No.1	Marks(5+5+5)
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- (a) Describe and differentiate the two most important network-layer functions in a datagram network? What is the 32-bit binary equivalent of the IP address 255.1.3.30?
- (b) Suppose there are three routers between a source host and a destination host. Ignoring fragmentation, an IP datagram sent from the source host to the destination host will travel over how many interfaces? How many forwarding tables will be indexed to move the datagram from the source to the destination? Draw the above scenario and describe in detail.
- (c) Find the shortest path using the Bellman Ford Algorithm from (S to all other nodes).

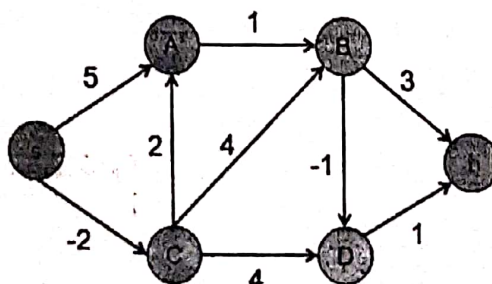


Figure 1



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Q.No.2

Marks(5+5+5)

- (a) How big is the MAC address space? The IPv4 address space? The IPv6 address space? Describe the Differences in IPv4 and IPv6.
- (b) Suppose the Information content of a packet is the bit pattern 1110 0110 1001 0101 and an even parity scheme is being used. What would the value of the field containing the parity bits be for the case of a two-dimensional parity scheme?
- (c) Consider the 5-bit generator, $G = 11011$, and suppose that D has the value 1010101010. What is the value of R ? Calculate sender side R and check correct CRC bits received by the receiver.

Q.No.3

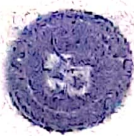
Marks(5+5)

- (a) Consider a router that interconnects three subnets: Subnet 1, Subnet 2, and Subnet 3. Suppose all of the interfaces in each of these three subnets are required to have the prefix 223.1.17/24. Also suppose that Subnet 1 is required to support at least 60 interfaces, Subnet 2 is to support at least 90 interfaces, and Subnet 3 is to support at least 12 interfaces. Provide three network addresses (of the form a.b.c.d/x) that satisfy these constraints.
- (b) What protocol is used for assigning IP address to a newly arrived node? Explain the whole process with the help of the diagram.

Q.No.4

Marks(5+5)

- (a) Draw an Infrastructure and infrastructure-less Wireless networks. What do you know about signal attenuation and Hidden terminal Problem in wireless networks?
- (b) What do you know about Symmetric key cryptography and Asymmetric key cryptography? Describe Active attacks and Passive attacks, How to create secure networks.



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Faculty	FoCAS	Department	CS & IT	Batch	2020F
Semester	5 th	Degree Program	BS(CS)	Course Instructor	Ms. Sana Ejaz
Course Code	CS-311	Course Title	Web Engineering	Credit Hours	2+0
Date of Examinations	14-02-2023	Timings	02.00 P.M -05:00 P.M	Shift	Evening

Timings: 3 Hours

Max Marks: 50

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Q.No.1	6 Marks
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Create a webpage having header, contents, navigation and footer sections. Header should contain the logo and/or name of your webpage and footer should have the copyrights statement. Display the contents and navigation of your own choice. Use HTML and CSS to design the above required Web page.

Q.No.2	10 Marks
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Suppose you are working in an organization and carry the responsibilities of "Web Developer". Your aim is to design the website for company's client who is based on certain requirements including tools and functional and non-functional requirements. The title of the website is "Online Store", which is a dynamic website and contains the information related to winter attires. Being the web developer give the answers of the following questions.

- What do you think about the requirements you will gather from your client?
- List down the steps through which a Web Engineer has to go through to develop a website.
- This particular web application lies in which website category.
- When you first meet the client what questions you will ask?
- Elaborate Requirement Elicitation Process for this scenario.
- Comment about its functional and nonfunctional requirements.
- Which website features you will develop at initial prototype?

Q.No.3	8 Marks
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- Write javascript program to find the largest among three numbers.
- Display a simple message "Web Engineering" on your Home webpage and when the user hovers over the message, an alert should be displayed with a message "Good Luck for your paper".
- Discuss the difference and advantages of JavaScript and jquery with suitable example.

Q.No.4	12 Marks
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- Create an XML file to hold data for the following.

Roll no	Name	Date of Birth	Height	Weight
2020F-CS-001	Ali	2-01-1999	5.4	52
2020F-CS-002	John	20-10-1992	4.2	48
2020F-CS-003	Ayesha	12-08-1997	5.7	65
2020F-CS-004	Talal	08-11-1995	6.2	78



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- b. Suppose a client needs a database design for his blog/website. Website has the following requirements. Design collection for MongoDB.
- Every post has the unique title, description and url.
 - Every post can have one or more tags.
 - Every post has the name of its publisher and total number of likes.
 - Every post has comments given by users along with their name, message, data-time and likes.
 - On each post, there can be zero or more comments.

Q.No.5	14 Marks
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- a. User created signup form (frontend) by using react framework nextjs and wants to connect with MongoDB. Explain the steps and command to connect nextjs with mongo db.
- b. Create a sign up form and validate it through javascript.

Login Validation:

- First and Last name should not contain any number.
- Use Regular Expression to validate Email.
- Password must have:
 - Capital letter
 - Smaller letter
 - Special Character
 - Digits from 0-9

Sign Up

First Name

Last Name

Email

Password

Already a member? Log In

Sign up



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Faculty	Computing & Applied Sciences	Department	CS&IT	Batch	2020F
Semester	5th	Degree Program	BS(CS)	Course Instructor	Quratulain khalid
Course Code	HS-401	Course Title	Technical Report Writing & Presentation Skills	Credit Hours	3+0
Date of Examinations	15-02-23	Timings	2.00 pm to 5.00 pm	Shift	Evening

Timings: 3 Hours

Max Marks: 50

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Q.No.1	Marks(3+2+5)
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- a) Shanzey has a strong background in social science research and evaluation. She manages staff that works on different projects within the agency. She is known as a problem solver and is extremely supportive of her staff. The managers are sensing that the staff is becoming overworked as everyone takes on increased responsibilities due to high staff turnover. They are having difficulty in appropriately allocating work to staff. (Shanze said that she had not received sufficient information from the finance department to complete the budgets. Keep in view the given scenario, if you were the manager/leader of that team and had to deliver bad news to your team, how would you have performed well with such a situation?
- b) How do you manage a conflict situation.
- c) Selecting a good team or building a good team which statement do you think is the quality of a leader and why? Justify your answer with an example.

Q.No.2	Marks(5+5)
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- a) How would you practice assertiveness as a professional when unnecessary or excessive favors are asked from you when you manage a team of 20 employees with multiple tasks



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to finish and deadline approaching near? Elaborate your response with the strategies that are essential to practice in such a scenario.

- b) Assertive people are selfish? Do you agree with the statement or not? Justify your answer with an example.

Q.No.3	Marks(7+3)
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- a) During an interview, as an interviewee we make number of blunders in front of the interviewer. If you go for an interview for a front desk position, how would you overcome your dressing issues and perform well during an interview.
- b) During an interview an employer harshly talk to you, then shouts on you and called it as your stress test do you think this type of test is appropriate for an interview? Explain with the justification that will you join that organization or not.

Q.No.4	Marks(7+3)
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- a) Microsoft made a shocking announcement that it was buying \$7.2 billion from Finnish mobile handset manufacturer Nokia, & merger aimed at building the mobile and Smartphone products of Microsoft. After the ink dried on the deal, the merger encountered even more difficulty, namely the difficulties of integrating workers from various cultures. According to the aforementioned case, how would Microsoft propose the significance of ideas during a table negotiation? Validate the situation through authentic steps when you negotiate with others in the organization to settle the agreement.
- b) What are the step to prepare for a negotiation.

Q.No.5	Marks(5+5)
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- a) List down and briefly explain 10 professional etiquettes that are important during a business meeting.
- b) What is the effective way to prepare for a presentation? What is an impromptu and extempore presentation.